Pd(II)-Catalyzed Coupling–Cyclization of 1,2-Allenyl Phosphonic Acid Monoesters with Allylic Halides and Their Application to the Synthesis of 1,3-Dihydro[2,1]benzoxaphosphole 1-Oxides

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Abstract: The palladium-catalyzed coupling–cyclization of monoesters of 1,2-allenyl phosphonic acids with allylic halides was studied. 4-Allyl-2,5-dihydro[1,2]oxaphosphole 2-oxides were smoothly afforded. Multisubstituted 1-ethoxy-1,3-dihydro[2,1]benzoxaphosphole 1-oxides were prepared from 3,4-diallyl-2,5-dihydro[1,2]oxaphosphole 2-oxides by a one-pot tandem reaction.

Key words: palladium-catalyzed, coupling–cyclization, monoester

Recently, phosphorus-containing heterocycles have drawn the attention of chemists because of their various biological activities[1] and catalytic properties.[2] Among these compounds, many heterocycles that include a phosphorus atom linked to an oxygen atom usually have important biological activities.[3] For synthesis of [1,2]oxaphosphole 2-oxides (compounds **IV** and **V** in Figure 1), some methodologies such as ring-closing metathesis (RCM),[4a] electrophilic cyclization,[4b,c] Reformatsky reaction,[3d] Dieckmann condensation,[4d,e] ester-exchange reaction,[3e,4f] or cyclization of 2-azidoallylphosphonates[4g] were studied. On the other hand, transition-metal-catalyzed cyclization reactions of allenes have been developed as efficient methods for the preparation of carbocycles as well as oxygen and nitrogen-containing heterocycles.[5] Although transition-metal-mediated cyclizations of functionalized allenes, such as those performed with Pd(0)[6] or Pd(II),[7] Ag(I),[8] Au(III),[9] Hg(II),[10] Ni(0),[11] or Ru(III),[12] have been well documented, the metal-catalyzed cyclization of phosphonic acids, which may provide an efficient pathway to the substituted phosphorus-containing heterocycles, have rarely been studied.

During the course of our studies on the chemistry of functionalized allenes, we have made some progress in the coupling–cyclization of 2,3-allenoic acids, 2,3-allenols or allenyl amines with allylic halides.[7e–h] Based on these results, we reasoned that a similar reaction of 1,2-allenyl phosphonic acid monoesters may provide an efficient route to 2,5-dihydro[1,2]oxaphosphole 2-oxides.

Our initial study began with ethyl (hepta-1,2-dien-3-yl)phosphonate (**1a**, Table 1). To our disappointment,

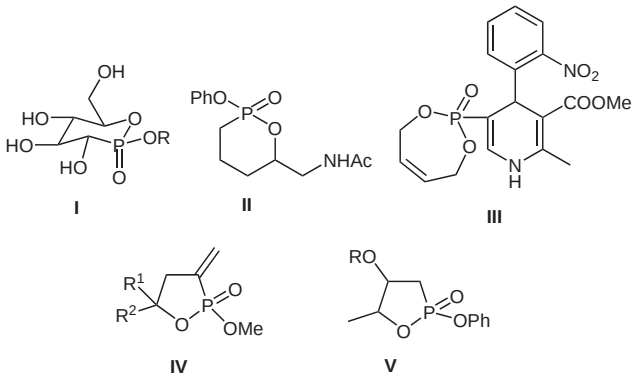


Figure 1

only O-allylation product **3a** was formed in the presence of 4 equivalents of K_2CO_3 under the catalysis of 10 mol% $PdCl_2(PPh_3)_2$ (entry 1, Table 1). However, the expected coupling–cyclization product **2a** was isolated in 14–55% yield in the presence of NaOH, LiOH, KOH, Cs_2CO_3 or $K_3PS_4{\cdot}3H_2O$ by using $PdCl_2(PPh_3)_2$ as catalyst. The reaction proceeded smoothly in the absence of any base in moderate yields (entries 2 and 3, Table 1). Then, after screening some reaction conditions, the reaction in DMF under the catalysis of 5 mol% $PdCl_2(PhCN)_2$ at 80 °C afforded **2a** as the single product in 72% yield (entry 4, Table 1). When the reaction was conducted in air, the yield of **2a** was lower (compare entry 4 with entry 6, Table 1). In terms of solvent effect, DMF is the best (compare entry 4 with entries 7–12, Table 1).

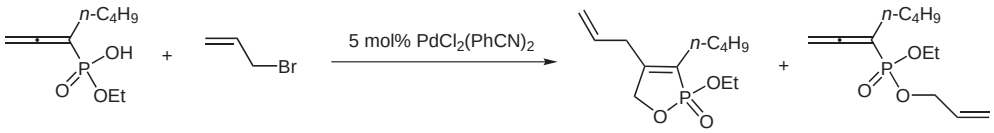
Subsequently, we studied the scope of this cyclization reaction under the standard conditions [5 mol% $PdCl_2(PhCN)_2$, DMF, 80 °C]. From the data listed in Table 2, it is obvious that the scope of the reaction is broad: R^1 can be H, alkyl, allyl, Bn or *t*-Bu and R^2, R^3 can be H, alkyl, $-(CH_2)_5-$. The 2,5-dihydro-[1,2]oxaphosphole 2-oxides **2** could be isolated in 42–75% yields.

1,2-Allenyl phosphonic acid monoester (**1f**) reacted smoothly with 2-methylallyl chloride to afford the expected product **2j** in 80% yield (Scheme 1). 4-(But-2′-enyl)-3-(*n*-butyl)-2-ethoxy-5,5-dimethyl-2,5-dihydro[1,2]oxaphosphole 2-oxide (**2k**) was prepared in 59% isolated yield with a *Z*/*E* ratio of 1:1 together with the cycloisomerization product **4f** in 37% yield by the reaction of **1f** with 3-chlorobuta-1-ene (Scheme 1).

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Table 1 Pd(II)-Catalyzed Coupling–Cyclization of Ethyl (Hepta-1,2-dien-3-yl)phosphonate **1a** with Allyl Bromide

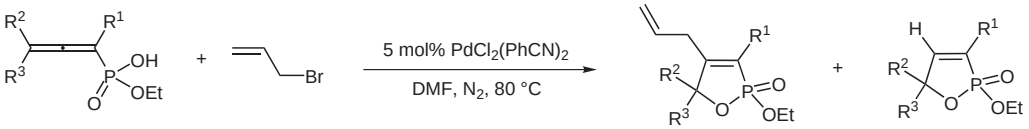


1a **2a** **3a**

Entry	Solvent	Temp (°C)	Time (h)	Yield of **2a** (%)
1[a]	DMF	70	48	0[b]
2[c]	DMF	70	48	55
3	DMF	70	3	65
4	DMF	80	2	72
5	DMF	90	1	68
6[d]	DMF	80	2	66
7	DMA	80	2	50
8	DMSO	80	2	32
9	CH_2Cl_2	reflux	24	61
10	THF	reflux	24	58
11	PhMe	80	3	26
12	MeCN	80	3	31

[a] The reaction was conducted in the presence of 4 equiv of K_2CO_3 using 10 mol% $PdCl_2(PPh_3)_2$ as the catalyst.
[b] Only **3a** (23%) was isolated.
[c] 5 mol% $PdCl_2(PPh_3)_2$ was used as the catalyst.
[d] The reaction was conducted in air.

Table 2 Pd(II)-Catalyzed Coupling–Cyclization of 1,2-Allenyl Phosphonic Acid Monoethyl Esters **1** with Allyl Bromide

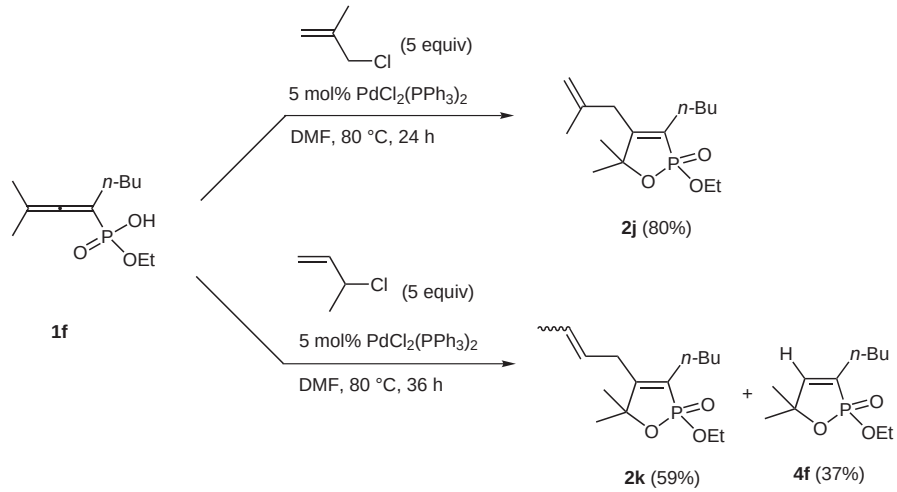


1 **2** **4**

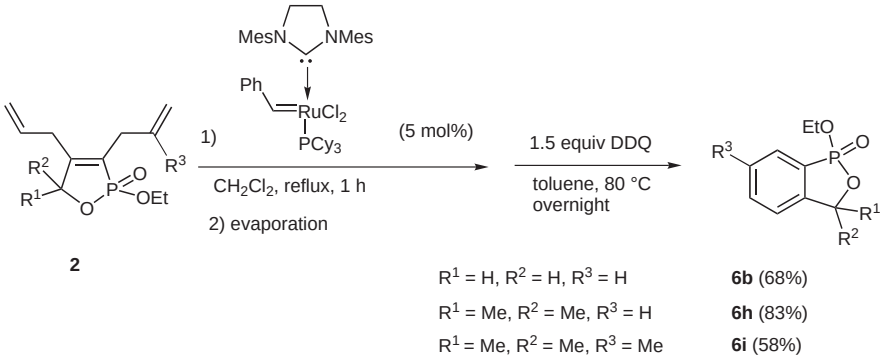
Entry	R^1	R^2	R^3	Compd **1**	Time (h)	Yield of **2** (%)
1	*n*-Bu	H	H	**1a**	2	72 (**2a**)
2	Allyl	H	H	**1b**	1	72 (**2b**)
3	*t*-Bu	H	H	**1c**	4	52 (**2c**)
4	Bn	H	H	**1d**	1	75 (**2d**)
5	H	Me	Me	**1e**	5	42 (**2e**)
6	*n*-Bu	Me	Me	**1f**	2	68 (**2f**)
7	*n*-Bu	$-(CH_2)_5-$		**1g**	5	67 (**2g**)
8	Allyl	Me	Me	**1h**	6	47 (**2h**)[a]
9	2-Methylallyl	Me	Me	**1i**	3	56 (**2i**)

[a] Compound **4h** was isolated in 12% yield.

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Scheme 1



Scheme 2

Then we decided to search for a novel and efficient procedure that could convert 3,4-diallyl-2,5-dihydro[1,2]oxaphosphole 2-oxides into 1,3-dihydro[2,1]benzoxaphosphole 1-oxides,[13] phosphorus analogues of 1(3*H*)-isobenzofuranone.[14] In the first place, 3,4-diallyl-2-ethoxy-2,5-dihydro[1,2]oxaphosphole 2-oxide (**2b**) was used as the starting material. With the Grubbs II catalyst,[15] 1-ethoxy-1,3,4,7-tetrahydro[2,1]benzoxaphosphole 1-oxide (**5b**) was afforded in good yield either in CH_2Cl_2 or toluene. Then DDQ (2,3-dichloro-5,6-dicyano-1,4-benzoquine) was applied as the oxidant[16] to afford 1-ethoxy-1,3-dihydro[2,1]benzoxaphosphole 1-oxide (**6b**) in 68% yield by a one-pot tandem reaction (Scheme 2). Under these conditions, some 3,3,6-substituted 1,3-dihydro[2,1]benzoxaphosphole 1-oxides were prepared in moderate to good yields (Scheme 2).

In conclusion, we have established a Pd-catalyzed coupling–cyclization of 1,2-allenyl phosphonic acid monoesters with allylic halides that affords the corresponding phosphorus-containing heterocycles, which can be converted into 1,3-dihydro[2,1]benzoxaphosphole 1-oxides when the 3,4-positions are substituted with two allylic groups. Further studies in this area are being conducted in our laboratory.

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